

Benchmarking Local Large Language Models for Structured Knowledge Extraction in Conversational Commerce

Ratomir Karlović¹ , Luka Sever¹ , Sandi Baressi Šegota¹ , Vedran Mrzljak²  and Ivan Lorencin¹ 

¹ Faculty of Informatics, Juraj Dobrila University of Pula, Zagrebačka 30, 52100 Pula, Croatia; ratomir.karlovic@unipu.hr (R.K.), luka.sever@unipu.hr (L.S.), sandi.baressi.segota@unipu.hr (S.B.Š.), ivan.lorencin@unipu.hr (I.L.)

² Faculty of Engineering Rijeka, University of Rijeka, Vukovarska 58, 51000 Rijeka, Croatia;

* Correspondence: vedran.mrzljak@riteh.uniri.hr;

Abstract

The integration of natural language understanding into conversational commerce requires robustly extracting structured intents from idiomatic text. While cloud-based models are highly capable, their latency, privacy risks, and operational costs necessitate efficient local alternatives. To address this, we empirically evaluate locally deployed, open-weights large language models (LLMs) on parsing multi-intent shopping cart commands into deterministic JSON schemas. We benchmark multiple models across minimal, extended, and few-shot prompting paradigms, utilizing a dual-axis framework to simultaneously measure extraction quality (operation F1-score, exact match, schema validity) and hardware performance (latency, GPU power consumption, VRAM usage). Our evaluation highlights a fundamental trade-off between schema compliance and computational overhead, demonstrating how advanced prompting techniques improve parsing accuracy while measurably impacting inference latency and power draw. Ultimately, this comprehensive profiling identifies optimal model configurations for resource-constrained environments, establishing a practical baseline for deploying cost-effective and energy-efficient LLMs in edge-based retail automation.

Keywords: LLM; Prompt Engineering; Natural Language Understanding; JSON Parsing; Structured Output; Model Performance; Conversational Agents; Information Extraction;

1. Introduction

The rapid evolution of Large Language Models (LLMs) has fundamentally transformed the landscape of Natural Language Processing (NLP), enabling advanced capabilities in reasoning, content generation, and semantic understanding [1]. While early applications focused primarily on open-ended text generation, modern production environments increasingly require LLMs to function as reliable semantic parsers capable of mapping unstructured user intent into rigorous, machine-readable formats such as JSON [2,3]. This capability is particularly critical in domain-specific applications, such as conversational commerce controllers, where valid schema adherence is a prerequisite for executing downstream backend tasks like inventory management and automated checkout workflows.

A primary challenge in deploying these systems within real-world retail architectures is balancing semantic parsing fidelity against operational and infrastructural constraints. While centralized, cloud-hosted proprietary models deliver high structural compliance,

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they introduce significant barriers regarding data privacy, volatile network latency, and high operational costs at scale [4]. Consequently, deploying open-weights models locally on edge or dedicated infrastructure has emerged as a compelling alternative to guarantee data sovereignty and predictable processing thresholds [5]. However, local deployments introduce an immediate engineering trade-off: smaller models frequently suffer from systemic structural failures, syntax corruption, or semantic omissions when tasked with generating strict object structures without heavy optimization.

In our previous work, we addressed these constraints by conducting a targeted empirical evaluation comparing Parameter-Efficient Fine-Tuning (PEFT) techniques against varied prompting complexities [1]. That study utilized a controlled synthetic dataset of 12,000 isolated, single-intent shopping directives (e.g., “Toss in 14 razors”) to parse specific entities into a simple three-attribute JSON schema. While those findings established foundational baselines showing that adapter-based methods or dense few-shot contexts could mitigate performance gaps across model scales, the evaluation was intentionally isolated from the linguistic noise and structural density characteristic of live human interactions.

To address these limitations, the current study significantly scales the structural and linguistic complexity of the semantic parsing task. Rather than processing isolated, single-intent inputs, we investigate the capacity of local open-weights models to unpack compound, multi-intent shopping cart commands compressed into a single utterance (e.g., “Toss in 14 razors in there, throw in 20 pies, load up with 19 coffees, and also nix a pair of bananas”). Parsing these compound structures requires an LLM to simultaneously determine disparate entity boundaries, map variable informal action synonyms (such as “load up” to an addition or “nix” to a removal), and resolve implicit numeric representations (such as “a pair” to an integer value of 2) within a single inference pass.

The primary aim of this research is to evaluate how distinct inference-time prompt engineering paradigms alter the parsing accuracy and hardware footprint of local models when processing these highly complex inputs. We implement a systematic benchmarking matrix crossing three prompting archetypes (Minimal, Extended, and Few-Shot) against a parameter-scale continuum of six local models: granite4:3b, llama3.2:3b, llama3.1:8b, mistral:7b, gpt-oss:20b, and gemma4:31b. Rather than analyzing extraction quality in isolation, we utilize a dual-axis framework that maps parsing metrics (operation F1-score, exact match rate, and schema validity) directly against physical hardware execution profiles (token latency, VRAM utilization, and GPU power consumption). This empirical setup uncovers the direct computational cost of prompt optimization, providing a data-driven baseline for selecting cost-effective and energy-efficient configurations for resource-constrained conversational agents.

2. Background and Related Work

The integration of Large Language Models (LLMs) into conversational agents has necessitated robust strategies for adapting general-purpose architectures to specialized, deterministic tasks. In complex e-commerce environments, agents must navigate noisy, heterogeneous user intents where implicit requirements frequently challenge standard generation capabilities. Tou et al. [6] established the ShoppingComp benchmark to evaluate autonomous agents across precise product retrieval and safety-critical decision-making, revealing a substantial human-AI capability gap where state-of-the-art models struggle with implicit requirement inference when translating vague intents into technical specifications. To evaluate the multi-step reasoning required to overcome these challenges, Wang et al. [7] introduced ShoppingBench, utilizing trajectory distillation to train a compact Qwen3-4B agent capable of complex budgeting tasks. Their findings highlighted that while smaller models can match closed-source baselines, attribute mismatching remains a

persistent bottleneck. To optimize agents for these multi-objective interactions, Cheng et al. [8] demonstrated that post-training Reinforcement Learning, specifically through Hierarchical Reward Modeling (HRM) and Dynamic Contrastive Policy Optimization (DCPO), significantly improves objective product correctness and operational tool efficiency without relying on generic reasoning paths.

To operate reliably in production environments, these models must exhibit high resilience to input perturbations and domain-specific constraints. Zhu et al. [9] evaluated LLM robustness against adversarial prompts, finding that while contemporary models suffer significant performance drops from word-level attacks, few-shot prompting and instruction fine-tuning are critical strategies for preventing attention divergence. Expanding on adaptation strategies, Wang et al. [10] demonstrated that fine-tuning open-weights models like Llama-3-8B and Qwen-7B for text classification effectively eliminates hallucinated outputs, reducing uncertainty rates to near zero compared to variable few-shot setups. In highly specialized domains, Dinh et al. [11] showed that translating natural language intents into machine-executable JSON or YAML configurations relies heavily on prompting design, with few-shot Chain-of-Thought (CoT) prompting drastically mitigating format errors. Similarly, in the clinical domain, Balasubramanian et al. [12] and Grothey et al. [13] demonstrated that advanced open-source models offer viable, privacy-preserving alternatives for structured feature extraction, though specialized strategies like Chain-of-Verification (CoV) are required to stabilize outputs upon 4-bit quantization. Extending structured extraction to scientific literature, Dagdelen et al. [14] utilized Low-Rank Adaptation (LoRA) and human-in-the-loop workflows to fine-tune Llama-2 for relation extraction, showing that LLMs automatically normalize complex chemical entities into structured JSON. Yang et al. [15] applied similar multimodal QLoRA tuning to the RRA-Logis framework for edge-computing, mapping noisy logistics documents directly to JSON schemas with 100% compliance via a confidence-gated decision mechanism. Evaluating such capabilities on a global scale, Ling et al. [16] introduced ResumeBench for multilingual parsing. Their results indicated that while enabling explicit JSON mode boosts syntactic fidelity, reasoning-distilled models surprisingly underperform compared to standard instruct counterparts, establishing models like Qwen2.5-14B as highly efficient deployment baselines.

2.1. Structured Output Generation and Benchmarking

To ensure reliable integration into transactional workflows, evaluating model proficiency in generating strictly constrained outputs has led to dedicated empirical frameworks. Yang et al. [17] introduced StructEval, a benchmark spanning diverse text-only and visual structures, demonstrating that generating valid structural outputs from natural language is significantly more difficult than simple format conversion. Addressing the fragility of standard "return valid JSON" prompting, Villota [18] argued that relying on probabilistic generation leads to high failure rates, advocating instead for constrained decoding at inference time to guarantee 100% schema adherence for text classification and retrieval tasks. Supporting this paradigm, Geng et al. [19] developed JSONSchemaBench to evaluate frameworks like Guidance and Outlines, revealing that optimized constrained decoding not only guarantees structural validity but can also accelerate the generation process by up to 50% and improve downstream reasoning tasks. To enforce this fidelity without massive parameter scaling, Shen et al. [20] introduced SLOT, utilizing a fine-tuned lightweight model as a dedicated post-processing layer, while Escarda-Fernandez et al. [21] demonstrated that QLoRA fine-tuning effectively overcomes zero-shot limitations when mapping natural language to custom JSON APIs on resource-constrained IoT devices.

When constrained decoding is not implemented, the choice of prompt style heavily dictates extraction success. White et al. [22] evaluated diverse prompt formats across frontier models, identifying a persistent trade-off: hierarchical formats like JSON provide high accuracy but incur higher token costs and latency compared to compact formats like CSV. Shorten et al. [23] explored these zero-shot limitations through the StructuredRAG benchmark, finding that task complexity rapidly degrades formatting reliability, though iterative prompt optimization (OPRO) can effectively bridge the performance gap for smaller models like Llama 3 8B. Pushing beyond static prompting, Lu et al. [24] proposed Schema Reinforcement Learning (SRL), utilizing a fine-grained validator and a "Thoughts of Structure" mechanism to significantly boost the schema compliance of Llama models beyond standard supervised fine-tuning. Taking an inverse approach, Shrimall et al. [25] introduced the PARSE framework, treating the JSON schema itself as an optimizable interface. By iteratively refining entity descriptions and validation rules, their system reduced extraction errors by up to 92% across multiple model scales, proving that optimizing the target structure mitigates inference latency and self-correction cycles.

In our previous research, we explored the "optimization dilemma" of structured parsing by comparing Parameter-Efficient Fine-Tuning (PEFT) against advanced prompting strategies for local LLMs [1]. Utilizing a dataset of 12,000 synthetic, single-intent shopping commands mapped to a rigid JSON schema, we found that LoRA-based adapters consistently yielded superior accuracy with zero inference-time overhead. However, the study also established that dense few-shot prompting provides a highly viable, training-free baseline that can mitigate performance gaps in smaller models, such as Llama 3.2 3B. The critical trade-off identified was that while prompt engineering requires no upfront training costs, it inherently increases long-term inference latency and computational resource allocation due to the extended token context required for multi-shot exemplars.

Based on the cumulative evidence from these studies, a critical gap remains in quantifying the exact hardware and latency costs of pure inference-time prompting when applied to highly complex, compound-intent inputs. While previous works have examined isolated intents or heavily orchestrated fine-tuning pipelines, the goal of this current study is to examine how distinct prompting archetypes influence both structured output reliability and physical hardware consumption. Guided by prior literature, the following hypotheses are considered:

- **H1:** Advanced prompt engineering paradigms, specifically those utilizing complex Few-Shot exemplars, will significantly improve multi-intent extraction accuracy and schema validity compared to minimal prompts across all evaluated model scales.
- **H2:** The structural and semantic accuracy gains achieved through extended context prompting will incur a measurable, non-linear increase in inference latency, peak VRAM allocation, and GPU power consumption.
- **H3:** Semantic mapping failures (e.g., incorrect value extraction or hallucinated intents) will persist more frequently in ultra-compact edge models ($\leq 3B$ parameters) despite advanced prompting, reflecting inherent limitations in compound reasoning compared to larger open-weights architectures.
- **H4:** Mid-tier local models (e.g., 7B to 8B parameters) utilizing advanced prompting will demonstrate the optimal cost-benefit ratio, bridging the structural capabilities of larger models while maintaining operational footprints suitable for edge deployment.

This paper consists of six main sections. Section 1 introduces the research problem and operational motivation. Section 2 reviews related work on local large language models and structured data extraction. Section 3 describes the empirical methodology, detailing the multi-intent dataset, prompt variants, evaluated model continuum, and hardware profiling metrics. Section 4 presents the experimental results across both parsing accuracy and

hardware efficiency axes. Section 5 synthesizes the findings, discusses the limitations of the evaluation, and outlines pathways for future research. Finally, Section 6 summarizes the core conclusions of the study.

3. Methodology

Materials and Methods should be described with sufficient details to allow others to replicate and build on published results. Please note that publication of your manuscript implicates that you must make all materials, data, computer code, and protocols associated with the publication available to readers. Please disclose at the submission stage any restrictions on the availability of materials or information. New methods and protocols should be described in detail while well-established methods can be briefly described and appropriately cited.

Research manuscripts reporting large datasets that are deposited in a publicly available database should specify where the data have been deposited and provide the relevant accession numbers. If the accession numbers have not yet been obtained at the time of submission, please state that they will be provided during review. They must be provided prior to publication.

Interventionary studies involving animals or humans, and other studies require ethical approval must list the authority that provided approval and the corresponding ethical approval code.

In this section, where applicable, authors are required to disclose details of how generative artificial intelligence (GenAI) has been used in this paper (e.g., to generate text, data, or graphics, or to assist in study design, data collection, analysis, or interpretation). The use of GenAI for superficial text editing (e.g., grammar, spelling, punctuation, and formatting) does not need to be declared.

4. Results and Discussion

We evaluate six open-weights models under three prompting regimes—*minimal*, *extended*, and *few-shot*—yielding 18 model-prompt configurations, each applied to the full set of 5000 held-out shopping-cart commands. These commands contain a mean of 3.64 atomic operations (18,175 gold operations in total), so a single utterance typically bundles several add/remove intents that must be jointly recovered. Following the dual-axis protocol introduced in the Methodology, we report extraction quality (Table 1) and computational cost (Table 2) for every configuration, and then analyse the quality-cost trade-off that the prompting regime induces (Figure 5). Throughout, we use the operation-level micro-averaged F1 score (Op-F1) as the primary quality scalar and an estimated energy per command, $E \approx \bar{P} \bar{t}$ (mean GPU power $\times$ mean latency), as the primary cost scalar.

4.1. Extraction Quality and the Role of the Prompting Regime

Table 1. Extraction quality by model and prompting regime (Min, minimal; Ext, extended; FS, few-shot). JSON and Schema denote the JSON- and schema-validity rates; Op-F1 is the operation-level micro-averaged F1 score; EM is the ordered full-command exact-match rate; Act, Prod, and Qty are the action-, product-, and quantity-field accuracies. All values are computed over the 5000-command evaluation set. The best Op-F1 per model is shown in bold.

Model	Prompt	JSON	Schema	Op-F1	EM	Act	Prod	Qty
Gemma4 31B	Min	1.000	1.000	0.951	0.833	0.969	0.995	0.984
	Ext	1.000	1.000	0.967	0.885	0.978	1.000	0.989
	FS	1.000	1.000	0.981	0.933	0.981	1.000	1.000
GPT-OSS 20B	Min	1.000	1.000	0.923	0.753	0.968	0.991	0.958
	Ext	0.999	0.999	0.944	0.819	0.981	0.994	0.967
	FS	1.000	1.000	0.975	0.915	0.995	0.999	0.980
Llama3.1 8B	Min	0.935	0.881	0.782	0.538	0.856	0.808	0.853
	Ext	0.998	0.976	0.801	0.568	0.929	0.860	0.949
	FS	0.997	0.995	0.950	0.841	0.977	0.966	0.988
Mistral 7B	Min	0.963	0.877	0.689	0.357	0.842	0.762	0.855
	Ext	0.982	0.865	0.504	0.281	0.871	0.559	0.887
	FS	0.993	0.976	0.942	0.804	0.950	0.963	0.973
Granite4 3B	Min	1.000	0.928	0.797	0.512	0.930	0.871	0.930
	Ext	1.000	0.931	0.790	0.516	0.921	0.859	0.946
	FS	1.000	0.997	0.937	0.807	0.984	0.958	0.987
Llama3.2 3B	Min	0.967	0.606	0.610	0.311	0.758	0.700	0.749
	Ext	0.970	0.637	0.526	0.229	0.681	0.581	0.754
	FS	0.890	0.822	0.813	0.566	0.796	0.810	0.826

Few-shot prompting yields the best Op-F1 for all six models, and the magnitude of the gain is inversely related to the model’s zero-shot capability. Relative to the minimal prompt, in-context demonstrations raise Op-F1 by only +0.031 for Gemma4 31B and +0.051 for GPT-OSS 20B, but by +0.140 (Granite4 3B), +0.168 (Llama3.1 8B), +0.203 (Llama3.2 3B), and +0.253 (Mistral 7B) for the smaller models. The exemplars thus act as a capability equaliser: models that are mediocre zero-shot parsers become competitive once shown the target format, whereas the largest models—already near ceiling—have little headroom to gain. Figure 1 visualises this compression of the quality gap between small and large models.

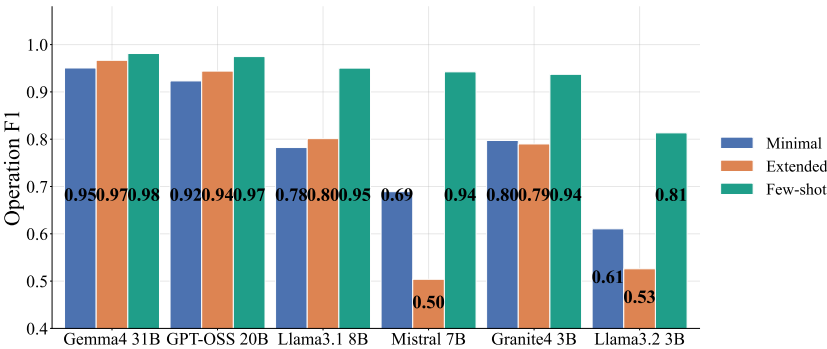


Figure 1. Operation-level F1 by model and prompting regime. Few-shot prompting (teal) yields the best accuracy for all six models and sharply compresses the gap between small and large models; the extended prompt (orange) is inconsistent, degrading Mistral 7B and the Llama3.2 3B small model relative to the minimal prompt (blue).

Decomposing quality into format adherence and content correctness clarifies the mechanism. Under the minimal prompt, several small models emit syntactically valid JSON that nonetheless violates the target schema: schema validity is only 0.928 (Granite4 3B),

0.881 (Llama3.1 8B), and 0.606 (Llama3.2 3B) despite JSON validity at or above 0.94. Few-shot demonstrations repair this format gap almost entirely—schema validity rises to 0.997, 0.995, and 0.822, respectively—indicating that the dominant failure mode of small models is structural (key naming, value typing, nesting) rather than semantic. Demonstrations specify the output contract by example rather than by description, which the smaller models follow more reliably than prose instructions. Figure 2 reports all seven quality metrics across models and regimes, making this format-versus-content decomposition visually explicit.

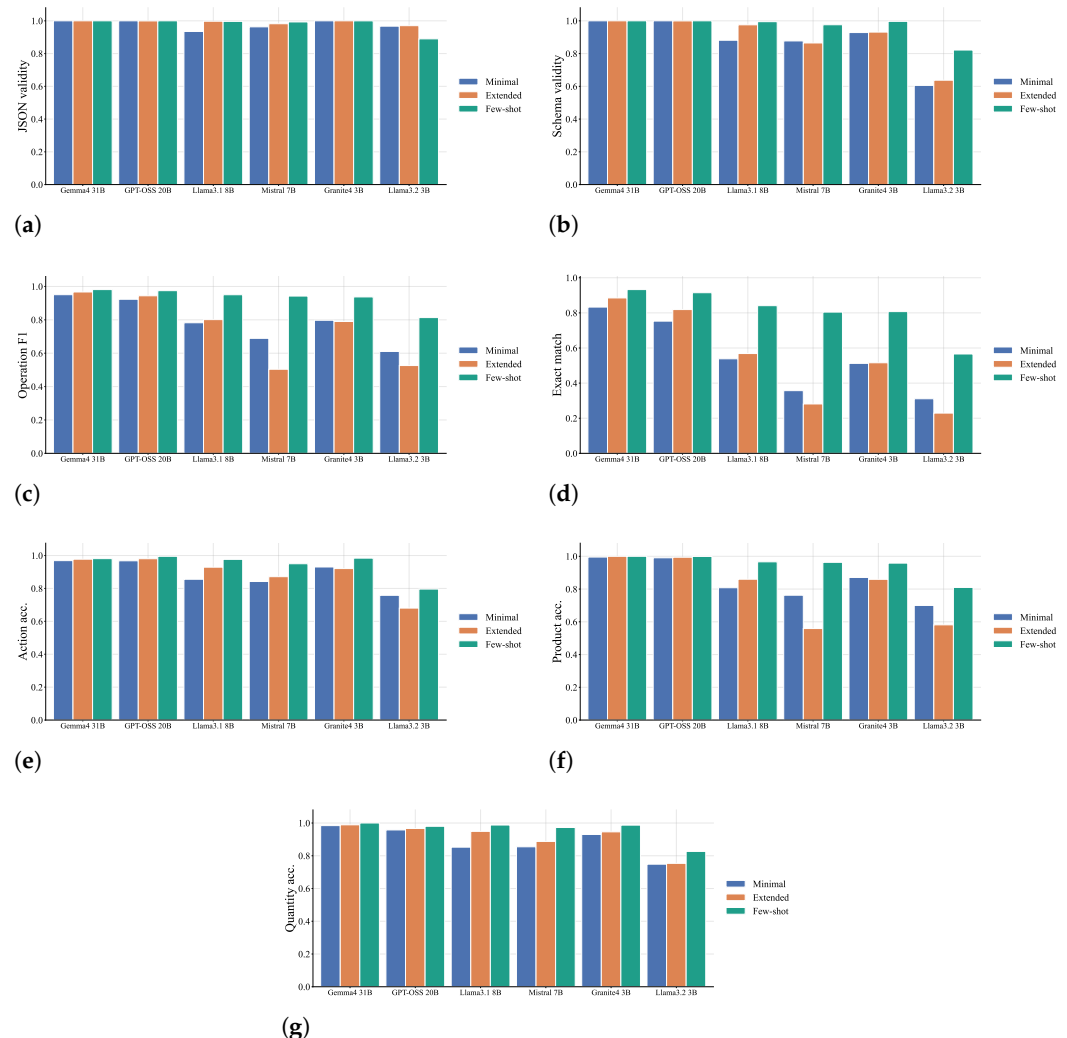


Figure 2. All quality metrics by model and prompting regime; each panel is a grouped bar chart over the six models (bars: minimal, extended, few-shot): (a) JSON validity, (b) schema validity, (c) operation F1, (d) exact match, (e) action accuracy, (f) product accuracy, and (g) quantity accuracy. Few-shot prompting (teal) is best across essentially every metric for the capable models, and the largest small-model gains appear on schema validity and exact match, confirming that the principal failure mode under weaker prompts is structural rather than semantic.

In contrast, the extended (instruction-rich) prompt is not a monotonic improvement and is frequently harmful. It helps only the most capable models—Gemma4 31B (+0.016), GPT-OSS 20B (+0.021), and Llama3.1 8B (+0.019)—while degrading Mistral 7B (−0.185) and Llama3.2 3B (−0.084), and leaving Granite4 3B essentially unchanged (−0.007). Manual inspection traces the Mistral regression to over-normalisation: the additional instructions induce the model to lemmatise product spans (“toothpastes” → “toothpaste”, “quinoa bags” → “quinoa bag”) and to misresolve idiomatic quantities (“a pair” → 1), collapsing

product-field accuracy from 0.762 to 0.559 even though the emitted JSON remains well-formed. Longer instructions therefore introduce degrees of freedom that smaller models resolve in unintended, ostensibly “helpful” ways that diverge from the verbatim-span ground truth. This is an important negative result: scaling prompt complexity without grounding it in exemplars can reduce extraction fidelity.

4.2. Computational Cost and the Decode-Bound Regime

Table 2. Computational cost by model and prompting regime. Lat_{med}, Lat_{mean}, and Lat_{p95} are latency percentiles (s); Thr is throughput (commands/s); Pwr is mean GPU power (W); E is the estimated energy per command (J, $E \approx \bar{P} \bar{t}$); P-tok and E-tok are the mean prompt- and output-token counts. Lower is better for all columns except throughput.

Model	Prompt	Lat _{med}	Lat _{mean}	Lat _{p95}	Thr	Pwr	E (J)	P-tok	E-tok
Gemma4 31B	Min	13.03	16.33	37.24	0.06	156	2555	146	600
	Ext	13.25	16.27	36.62	0.06	156	2545	197	599
	FS	10.10	11.71	19.87	0.08	156	1820	436	426
GPT-OSS 20B	Min	2.46	2.54	3.90	0.36	124	315	196	345
	Ext	2.44	2.52	3.95	0.36	124	313	248	341
	FS	2.39	2.51	3.89	0.37	126	315	455	339
Llama3.1 8B	Min	0.72	0.71	0.98	1.06	141	101	138	79
	Ext	0.64	0.66	0.93	1.12	141	94	191	72
	FS	0.58	0.56	0.71	1.26	140	79	402	57
Mistral 7B	Min	0.61	0.59	0.84	1.24	146	87	148	88
	Ext	0.58	0.56	0.78	1.30	147	82	200	82
	FS	0.50	0.49	0.64	1.42	143	70	467	70
Granite4 3B	Min	0.65	0.61	0.82	1.21	119	73	136	98
	Ext	0.53	0.52	0.81	1.35	118	62	189	81
	FS	0.41	0.39	0.51	1.64	118	46	400	56
Llama3.2 3B	Min	0.51	0.53	0.76	1.34	123	65	148	90
	Ext	0.50	0.49	0.63	1.42	124	61	201	80
	FS	0.40	0.39	0.48	1.67	122	47	412	56

The cost profile is governed by decoding rather than prefill. Although few-shot prompts are three- to fourfold longer on the input side (prompt tokens rise from ~140 to ~400–470), they consistently reduce wall-clock latency because the demonstrations suppress verbose generation: mean output length falls (e.g., 98 → 56 eval tokens for Granite4 3B and 600 → 426 for Gemma4 31B), and decode dominates latency in autoregressive inference. Consequently, few-shot mean latency is 17.7–35.6% lower than minimal for five of the six models (Granite4 3B –35.6%, Gemma4 31B –28.3%, Llama3.2 3B –27.2%, Llama3.1 8B –21.2%, Mistral 7B –17.7%), with GPT-OSS 20B effectively flat (–1.4%). The conventional intuition that richer prompts cost more compute is thus inverted here: by constraining the output, exemplars make inference simultaneously more accurate and faster. Figure 3 contrasts the prompt and output token counts that drive this effect.

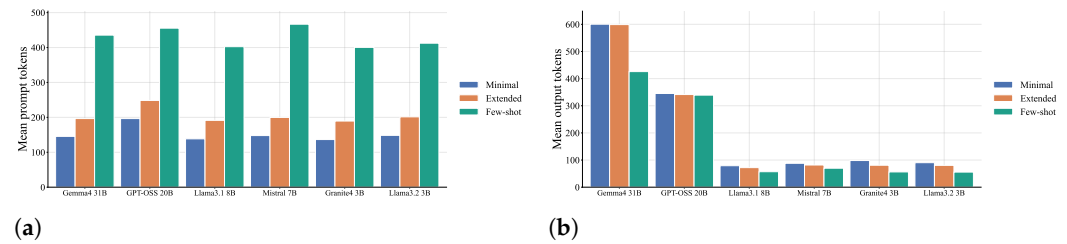


Figure 3. Mean prompt (input; **a**) and output (decoded; **b**) token counts per command, by model and prompting regime. Few-shot prompts are markedly longer on the input side (**a**) yet *shorter* on the output side (**b**): the demonstrations elicit concise, minified JSON and suppress verbose generation. Because decoding dominates latency, this reduction in output tokens is the mechanism behind the lower few-shot latency and energy reported in Table 2.

GPU power draw is essentially model-bound rather than prompt-bound—within each model it varies by only a few watts across regimes (e.g., ~156 W for Gemma4 31B, ~124 W for GPT-OSS 20B, ~118 W for Granite4 3B)—so the estimated energy per command tracks latency. Few-shot prompting therefore reduces per-command energy by 19–36% for the capable models, with the smallest models reaching 46 J (Granite4 3B) and 47 J (Llama3.2 3B) per command versus 315 J (GPT-OSS 20B) and 1820 J (Gemma4 31B). Energy spans a factor of roughly 40 across the frontier—a far wider dynamic range than quality—which is the crux of the deployment trade-off analysed in Section 4.3. Figure 4 summarises all eight cost metrics across the 18 configurations.

We report mean VRAM occupancy in the underlying logs for completeness but caution against ranking models by it: the measured footprint does not track parameter count (the 20B GPT-OSS model occupies only ~16 GB, less than the 3B Llama3.2 at ~17 GB or the 8B Llama3.1 at ~22 GB), which indicates that the figure reflects server-level allocator reservation and key-value-cache pre-allocation under the serving runtime rather than the minimal model footprint. Absolute VRAM values should accordingly be read as an upper bound on residency for this deployment, not as an intrinsic model property.

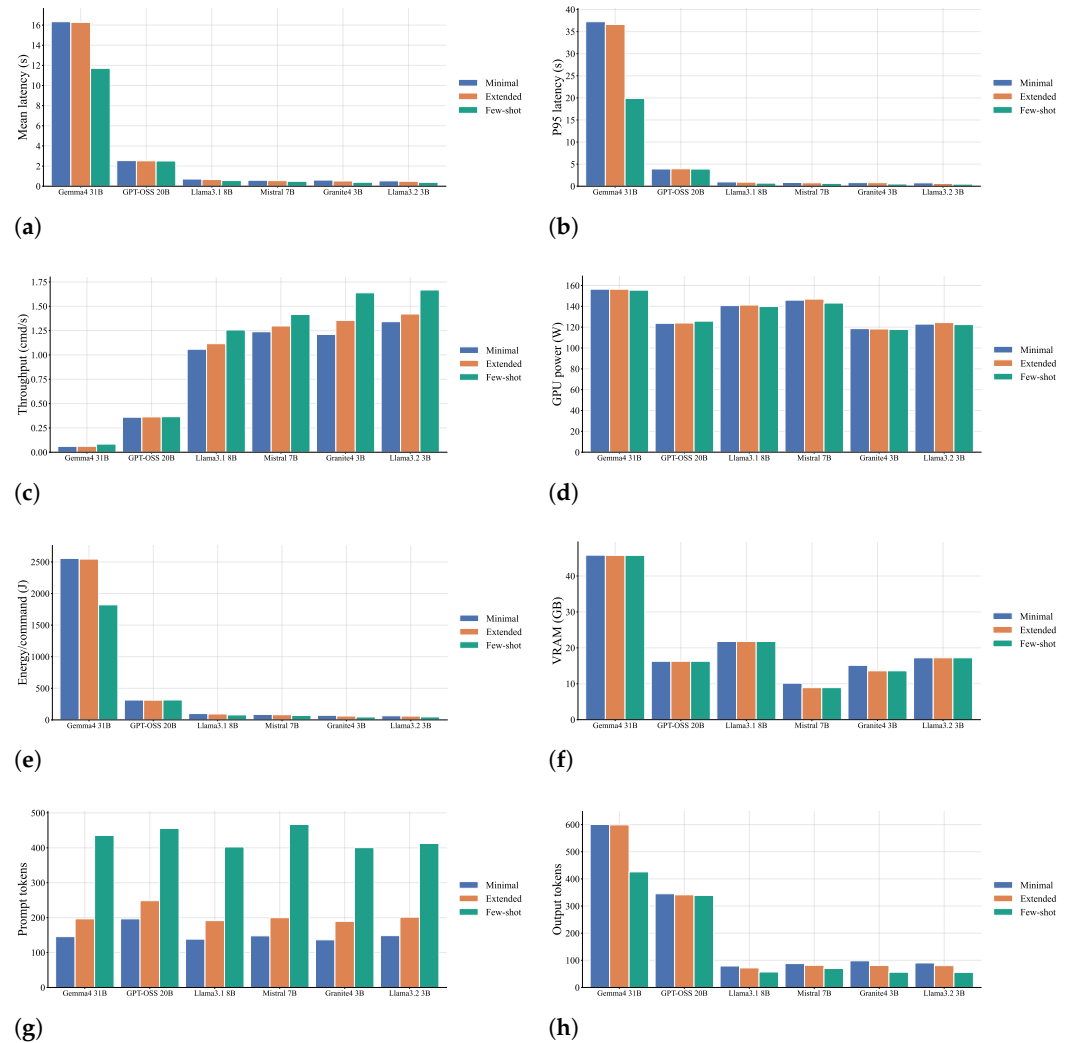


Figure 4. All cost metrics by model and prompting regime: (a) mean latency, (b) p95 latency, (c) throughput, (d) mean GPU power, (e) estimated energy per command, (f) VRAM occupancy, (g) mean prompt tokens, and (h) mean output tokens. Latency, energy, and throughput improve under few-shot prompting for every model, whereas GPU power is essentially model-bound. The latency and energy panels are dominated by Gemma4 31B, reflecting the $\sim 40\times$ cost dynamic range across the frontier. VRAM reflects serving-runtime reservation rather than the minimal model footprint (see text).

4.3. The Quality–Cost Trade-off

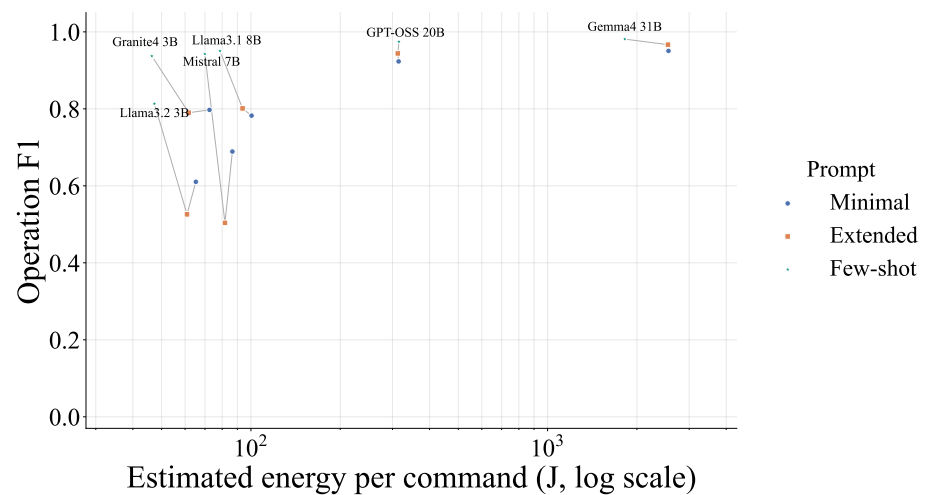


Figure 5. Quality–cost trade-off across all 18 configurations. Each point is one model–prompt configuration; thin grey lines connect the three regimes of a model. The horizontal axis is the estimated energy per command on a logarithmic scale. For every model the few-shot configuration (star) lies above and to the left of its minimal (circle) and extended (square) counterparts—simultaneously more accurate and more energy-efficient—so the operative trade-off is across model scale along the few-shot frontier rather than across prompts.

Figure 5 plots every configuration in the energy–quality plane and reveals the central finding of this study: for capable models, few-shot prompting is Pareto-improving rather than Pareto-trading—it moves each model up (higher F1) and to the left (lower energy) at once. The few-shot points dominate their own minimal and extended counterparts for all six models. The genuine trade-off is therefore not between prompts but across model scale: along the few-shot frontier, quality and cost are positively coupled, and the practitioner’s decision reduces to selecting an operating point on that frontier. Three operating points are practically distinct:

- **Maximum accuracy.** Gemma4 31B (few-shot) attains the highest fidelity (Op-F1 0.981, exact match 0.933) but at 1820 J and 11.7 s per command—appropriate only where accuracy is paramount and latency and energy are unconstrained, such as offline or server-side processing.
- **Balanced.** GPT-OSS 20B (few-shot) reaches Op-F1 0.975 at 315 J and 2.5 s, a 5.8× energy reduction for a 0.6-point F1 loss relative to Gemma4 31B.
- **Efficiency frontier.** Granite4 3B (few-shot) achieves Op-F1 0.937—95.5% of Gemma4 31B’s quality—at 46 J and 0.39 s, that is, roughly 40× less energy and 30× lower latency. Llama3.1 8B (0.950, 79 J) and Mistral 7B (0.942, 70 J) offer marginally higher accuracy at modestly higher cost.

Quality saturates rapidly with scale once few-shot prompting is applied: moving from a 3B to a 31B model improves Op-F1 by only ~ 0.044 while increasing energy by a factor of roughly 40. For the resource-constrained, edge-oriented retail setting that motivates this work, a 3–8B model with few-shot prompting is therefore the rational default, reserving 20–31B models for accuracy-critical or server-side deployments. Crucially, this recommendation is contingent on the prompt: the same small models are not viable under minimal or extended prompting (Op-F1 0.50–0.80), so the deployment win arises from the prompt–model pairing rather than from the model alone. The same Pareto structure holds when mean latency replaces energy on the cost axis (Figure 6), confirming that the ranking of configurations is robust to the choice of cost metric.

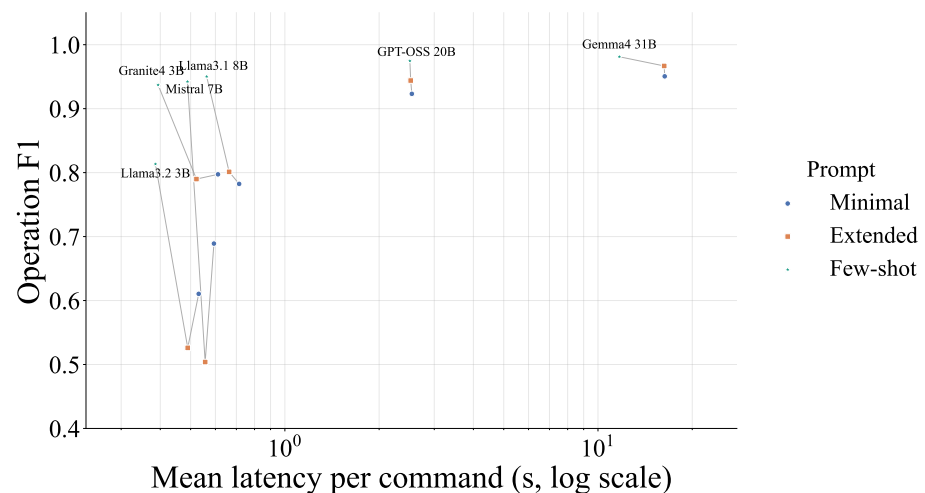


Figure 6. Quality–cost trade-off with mean latency per command (log scale) as the cost axis, complementing Figure 5. Markers denote the prompting regime and thin lines connect the three regimes of each model. The few-shot frontier again lies up and to the left and the model ordering is preserved, confirming that the conclusions are robust to whether cost is measured as latency or energy.

4.4. Error Analysis

Field-level accuracies localise the residual errors. Product-span extraction is the hardest field for every model that makes appreciable errors: under the minimal prompt the 3–8B models score only 0.70–0.87 on the product field, below both their action (0.76–0.93) and quantity (0.75–0.93) accuracies, reflecting sensitivity to surface form such as pluralisation; quantity resolution is additionally stressed by idiomatic numerals (“a pair”, “half a dozen”, “a dozen”) that must be mapped to integers. For the near-ceiling large models the three fields all exceed 0.95 and their differences are negligible. The gap between operation-level F1 and whole-command exact match is informative: even Gemma4 31B (few-shot) scores 0.981 Op-F1 but 0.933 exact match, because a command counts as correct only if all of its 3.64 operations on average are simultaneously correct, so per-operation errors compound multiplicatively at the command level. Exact match is consequently the more stringent and deployment-relevant metric for transactional settings, where a single misparsed item corrupts the entire cart update.

5. Limitations and Future Work

6. Conclusions

This section is not mandatory, but can be added to the manuscript if the discussion is unusually long or complex.

Author Contributions: For research articles with several authors, a short paragraph specifying their individual contributions must be provided. The following statements should be used “Conceptualization, X.X. and Y.Y.; methodology, X.X.; software, X.X.; validation, X.X., Y.Y. and Z.Z.; formal analysis, X.X.; investigation, X.X.; resources, X.X.; data curation, X.X.; writing—original draft preparation, X.X.; writing—review and editing, X.X.; visualization, X.X.; supervision, X.X.; project administration, X.X.; funding acquisition, Y.Y. All authors have read and agreed to the published version of the manuscript.”, please turn to the [CRediT taxonomy](#) for the term explanation. Authorship must be limited to those who have contributed substantially to the work reported.

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Abbreviations

The following abbreviations are used in this manuscript:

LLMs	Large Language Models
NLP	Natural Language Processing
PEFT	Parameter-Efficient Fine-Tuning
LoRA	Low-Rank Adaptation
IA ³	Infused Adapter by Inhibiting and Amplifying Inner Activations
ICL	In-Context Learning
JSON	JavaScript Object Notation
SFT	Supervised Fine-Tuning
F1	F1 Score
GPU	Graphics Processing Unit
DDP	Distributed Data Parallel
MLP	Multi-Layer Perceptron
bf16	BFloat16 Precision Format
tf32	TensorFloat-32

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